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**Sentiment Analysis of Drug Reviews Using
Natural Language Processing Techniques**

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1. Introduction

1.1 Project Overview

This project was carried out in two main parts. The first part focused on extracting and analyzing sentiments in drug reviews through three key techniques: opinion mining, aspect-based sentiment analysis (ABSA), and sentiment classification. Opinion mining and ABSA were used to identify specific opinion-aspect pairs and sentiment labels, offering more context beyond just overall polarity. For classification, traditional machine learning models (Support Vector Machine and Logistic Regression with TF-IDF features) were first used to set a baseline, followed by a fine-tuned DistilBERT Transformer model. To address class imbalance and overfitting, a weighted loss function and early stopping were implemented. The Transformer model achieved a final weighted F1-score of 80.5%, showing a clear improvement over the baseline methods.

The second part of the project focused on understanding what drives negative user experiences by analyzing the most common terms and patterns found in negative reviews. This helped surface recurring issues such as side effects, lack of effectiveness, or mismatched expectations—adding a layer of explainability to the classification outcomes. Together, both parts of the project aimed to go beyond basic sentiment prediction by uncovering the "why" behind user dissatisfaction, offering more meaningful insights from the data.

2. Problem Statement

With the rise of online health forums and review platforms, there's been an overwhelming growth in patient-generated drug reviews that contain useful information about medication experiences—including effectiveness, side effects, and overall

satisfaction. While this content is rich in insight, its unstructured format and large volume make manual analysis not only impractical, but also inefficient and prone to inconsistency.

This project addresses the challenge by applying Natural Language Processing (NLP) techniques to automatically classify these reviews into positive, neutral, or negative sentiments. By using both traditional machine learning and transformer-based models on real-world data, the goal is to help healthcare providers, researchers, and patients better understand public sentiment toward medications—faster, and with greater accuracy.

3. Literature Review

3.1 Sentiment Analysis of Drug Reviews Using Machine Learning Techniques

This paper dives into sentiment analysis of user-generated drug reviews, aiming to tap into the wealth of patient feedback about medication experiences. With data like effectiveness, side effects, and overall satisfaction increasingly available online, the study highlights the importance of leveraging such insights to guide healthcare decision-making, improve products, and enhance patient care. The goal is to classify reviews into positive, neutral, or negative sentiment categories using advanced machine learning (ML) and deep learning (DL) techniques.

The authors, Nair et al. (2024) used a dataset of 5,170 drug reviews sourced from WebMD, which underwent thorough preprocessing. This involved cleaning up noise, tokenizing, and lemmatizing to ensure the text was ready for analysis. They relied on pre-trained models like BERT, BioBERT, and SciBERT to generate embeddings—dense numerical representations capturing the context and meaning of words in the reviews. These embeddings were then fed into classifiers like Random Forest, Support Vector Machines (SVMs), Decision Trees, and Recurrent Neural Networks (RNNs), and evaluated using precision, recall, and F1 scores.

Key findings showed that embedding models significantly improved sentiment classification, with Random Forest delivering the best performance among ML algorithms, while RNNs excelled in handling more complex sentiment patterns. The study addressed

common issues like data imbalance using techniques like SMOTE to ensure fair representation of sentiment categories during training.

The research stands out for its focus on pre-trained models tailored for healthcare-related text, proving their effectiveness in extracting meaningful patterns from drug reviews. However, the authors acknowledge limitations like the modest size of the dataset and suggest future research directions—such as training models on larger datasets and experimenting with transformer-based architectures like BERT to further refine sentiment classification.

In conclusion, this paper highlights how advanced AI tools can be used to process patient feedback and inform healthcare decisions. Its findings have practical implications for pharmaceutical companies, healthcare providers, and even policymakers, offering new ways to improve drug monitoring, detect adverse reactions, and customize treatments based on real-world patient experiences.

3.2 Deep Learning-Based Method for Sentiment Analysis for Patients' Drug Reviews

This research by (Al-Hadhrami et al., 2024) takes an in-depth look at the potential of deep learning techniques for analyzing patients' drug reviews, a growing source of information that reflects real-world healthcare experiences. By focusing on datasets from Drug Review and WebMD, the authors explore the efficacy of two state-of-the-art models: bidirectional LSTM (Bi-LSTM) and a hybrid Bi-LSTM-CNN architecture. Their goal is to evaluate how well these models can classify sentiments from reviews and whether incorporating GloVe word embeddings—a pre-trained semantic representation technique—can enhance this process.

To ensure high-quality data input for the models, the study implements thorough preprocessing steps. These include tokenization, lemmatization, removal of stop words, and other cleaning methods designed to prepare the text for analysis. The hybrid Bi-LSTM-CNN model emerges as a standout performer, combining CNN's ability to detect local patterns with Bi-LSTM's capacity to capture the contextual relationships in sequential data.

This enables the model to deliver robust sentiment classification results, with the hybrid architecture achieving a remarkable 96% accuracy on one of the datasets. The addition of GloVe embeddings proves beneficial, enriching the models' capacity to interpret nuanced expressions, although its impact varies depending on the characteristics of the dataset.

One of the paper's key contributions is its comparative analysis, which highlights the advantages of hybrid architectures over standalone approaches. It demonstrates how the combination of CNN and Bi-LSTM layers can overcome the limitations of each individual method, offering a more comprehensive understanding of complex text data. At the same time, the authors are candid about the challenges they faced, including issues with data imbalance and the relatively limited scope of available datasets. To address these limitations, they propose future research into transformer-based models like BERT, which could further refine performance and expand the scope of analysis to include a wider range of healthcare scenarios.

This study goes beyond academic inquiry by emphasizing the practical implications of its findings. The demonstrated potential of AI-driven sentiment analysis has significant applications in healthcare—enabling providers to better understand patient experiences, identify adverse drug reactions, and refine drug recommendations. By leveraging these insights, healthcare organizations and policymakers could make more informed decisions that ultimately lead to improved patient outcomes.

In summary, this work underscores the value of integrating advanced AI techniques into sentiment analysis for healthcare. By combining rigorous methodology, innovative model architectures, and practical applications, it represents a step forward in how user-generated content can be harnessed to drive meaningful changes in the healthcare industry. The study not only raises the bar for future research but also provides a foundation for exploring new approaches to understanding and improving patient care.

3.3 Data-Driven Solution to Identify Sentiments from Online Drug Reviews

This paper by (Haque et al. ,2023). dives into sentiment analysis (SA) of online drug reviews, mixing machine learning (ML) and deep learning (DL) techniques to classify reviews as positive, neutral, or negative. It tackles big challenges like cleaning noisy data and dealing with imbalanced datasets, while also setting out to outperform earlier efforts that struggled with accuracy and scalability. Essentially, it's about finding smarter, more effective ways to extract meaningful insights from drug reviews.

The study combines five ML models (like Random Forest and Logistic Regression) and four DL models (like LSTM and Bi-LSTM), testing them with different feature extraction techniques, such as Count Vectorizer (CV) and GloVe embeddings. What stands out is that the Random Forest model trained on the CV achieves a solid 96.65% accuracy, outperforming other ML methods and raising the bar. However, the Bi-LSTM trained on GloVe embeddings goes even further, achieving a remarkable 97.40% accuracy and an F1 score of 97.42%. This is thanks to its ability to process text both forward and backward, making it better at grasping context and nuances in the reviews.

What makes this research more than just academic is its practical application. The team developed a web application powered by the top-performing RF model, designed to classify drug reviews automatically. This tool has real-world implications—it could help healthcare professionals and pharmaceutical companies detect adverse drug reactions sooner, improve drug monitoring, and support better decision-making.

While the results are impressive, the authors acknowledge there's room for growth. They suggest exploring advanced models like transformers (think BERT or GPT) to address challenges such as data imbalance and refining predictions further. Despite these challenges, the study provides meaningful insights into how user-generated content can be used to enhance healthcare systems, making it a valuable contribution to the field. It's a great example of tech being used to simplify complex problems and make a real difference.

3.4 Drugs Rating Generation and Recommendation from Sentiment Analysis of Drug Reviews using ML

In their 2020 paper, Hossain et al., (2020) proposed a drug recommendation system that leverages sentiment analysis of user-generated drug reviews. Their approach aims to address the growing need for personalized healthcare recommendations as online health information proliferates. The authors highlight a significant gap in recommendation systems research—while sentiment analysis has been extensively applied to products, movies, and restaurants, its application in healthcare and specifically drug reviews remains relatively unexplored. Their work contributes to filling this gap by developing a decision-making support platform to help patients make more informed choices in drug selection.

The proposed framework comprises five main modules: data preprocessing, feature extraction, rating generation, dictionary sentiment analysis, and recommendation. The authors employ several machine learning algorithms to convert unstructured drug review text into numerical ratings, testing Decision Tree (DT), K-Nearest Neighbors (KNN), and Linear Support Vector Classifier (LinearSVC) approaches. Their methodology involves careful data preprocessing, including tokenization, stop word removal, handling negative adjectives, and lemmatization.

For feature extraction, they implement unigrams, bigrams, and trigrams models to capture nuanced sentiment expressions like "not very good." The authors utilize a dataset from the UCI Machine Learning Repository derived from Drug.com and Druglib.com, which includes user ratings on a scale of 1-10 (converted to 1-5 for their purposes). Their experimental results show that LinearSVC outperforms the other algorithms with 83.08% accuracy compared to DT (76.79%) and KNN (55.41%).

An interesting aspect of their work is the implementation of dictionary sentiment analysis using the Harvard emotional dictionary to overcome limitations of sentiment packages built on movie review data. This demonstrates thoughtful consideration of domain-specific challenges in healthcare sentiment analysis.

The recommendation model filters drugs based on condition descriptions and recommends appropriate medications for specific health conditions. The paper provides example recommendations for conditions like ADHD and abdominal distension. While the paper makes valuable contributions, it has some limitations.

The authors acknowledge that future work could incorporate more linguistic rules, phrase-level sentiment analysis, and advanced techniques like tensor factorization or deep learning to further enhance accuracy and reliability. Overall, this work represents a promising step toward leveraging user-generated content to improve drug recommendation systems, with potential applications for both patients and healthcare providers. The combination of sentiment analysis with traditional recommendation systems offers a more nuanced approach to matching patients with appropriate medications.

4. Methodology

4.1 Data Collection

The dataset used in this project was sourced from Kaggle's Drug Review Dataset, which contains over 100,000 user-submitted reviews of various medications. Each entry includes features like the drug name, condition, user review and rating (1-10), providing a comprehensive foundation for sentiment analysis and opinion mining tasks.

4.2 Data Preprocessing

To ensure the quality and consistency of the data, a multi-stage preprocessing pipeline was implemented. The process involved the following sequential steps:

- **Sentence Segmentation:** The raw review text was first segmented into individual sentences using NLTK's `sent_tokenize`.
- **Word Tokenization:** Each sentence was then broken down into a list of individual words or tokens via `word_tokenize`.
- **Case Folding:** All tokens were converted to lowercase to ensure uniformity across the dataset.
- **Punctuation Removal:** All punctuation characters were identified and removed from the list of tokens.
- **Stopword Removal:** Common, low-value English words (e.g., "the", "a", "in") were filtered out from the token list using NLTK's standard stopwords collection.

- **Lemmatization:** The token list was processed with the WordNetLemmatizer to reduce each word to its base dictionary form (e.g., "studies" becomes "study"), consolidating their meaning.
- **Raw Text Cleaning for Transformer Model:** A separate, comprehensive cleaning function was applied to the original raw text to prepare it for the DistilBERT model. This function performed several actions:
 - Converted HTML entities (e.g., ') to standard characters.
 - Removed noise such as escaped characters, line breaks, URLs, and email addresses.
 - Normalized excessive punctuation (e.g., !!! to !) while preserving essential punctuation marks.
 - Corrected spacing issues and removed leading/trailing whitespace.
 - Expanded common abbreviations (e.g., mg to milligrams) to provide clearer context.

4.3 Feature Engineering

To prepare the review texts for machine learning models, the Term Frequency-Inverse Document Frequency (TF-IDF) technique was applied. This technique creates a meaningful numerical representation by evaluating a word's importance within a review relative to the entire collection of reviews. The TfidfVectorizer was configured to optimize the feature set by limiting the vocabulary to the top 5,000 most frequent terms. It was also set to ignore words that were either too rare (appearing in only one review) or overly common (appearing in more than 80% of all reviews), ensuring the models focused on the most discriminating terms. The resulting numerical matrix served as the input feature set for the SVM and Logistic Regression models.

4.4 Opinion Mining

To move beyond document-level sentiment and extract specific opinions, a custom Opinion Miner was developed. This initial approach used a combination of linguistic

parsing and rule-based patterns to identify not just the sentiment but also the specific target of that sentiment. The methodology involved:

- **Explicit Opinion Extraction:** Identifying sentiment by checking each word against custom-defined opinion lexicons (e.g., pos_words, neg_words), with VADER used as a fallback.
- **Implicit Opinion Extraction:** Using regular expressions to find phrases where sentiment is implied (e.g., "symptoms disappeared").
- **Target Identification:** Once an opinion word was found, spaCy's dependency parsing was used to find its grammatical target, such as an adjective's corresponding noun ("<u>effective</u> medication") or a verb's direct object ("<u>love</u> this drug").

Following the extraction process on a sample of 30,000 reviews for faster processing, a comprehensive analysis was conducted to aggregate the findings. This analysis included calculating overall statistics, such as the total number of opinions extracted and the percentage of reviews containing at least one opinion. The distribution of sentiment polarity (positive vs. negative) and opinion type (explicit vs.implicit) was also examined. Finally, the aggregated data was used to rank and identify the most frequent positive and negative opinion words, the most commonly discussed targets, and the most prevalent opinion-target pairs, providing a clear framework for understanding patient feedback. The results/findings of this part will be further elaborated in section 7.

4.5 Aspect Based Sentiment Analysis (ABSA)

To complement the rule-based miner, a more exploratory **Aspect-Based Sentiment Analysis (ABSA)** pipeline was implemented. This approach focused on first discovering relevant topics (aspects) directly from the data and then assigning sentiment to them. The methodology was as follows:

- **Aspect Extraction:** Potential aspects were automatically identified from the text using spaCy to extract meaningful noun phrases (e.g., "side effects") and descriptive adjective-noun pairs (e.g., "bad cramps").

- **Sentiment Assignment:** For each discovered aspect, the system located every sentence in the review where the aspect was mentioned. The VADER sentiment analyzer was then used to calculate a sentiment score for each of those sentences, which were averaged to determine the final sentiment for that aspect within the review.

Following the extraction process, the results were aggregated to identify high-level trends. The analysis involved counting the total mentions for every unique aspect discovered across all reviews to rank them by frequency. For the most prominent aspects, the overall sentiment polarity was then analyzed by calculating the percentage distribution of positive, negative, and neutral mentions. This provided a clear, data-driven summary of the most important topics to patients and the collective sentiment associated with them. The results/findings of this part will be further elaborated in section 7.

4.6 Model Building

- **Machine Learning Models:** Support Vector Machine (SVM) and Logistic Regression (LR) classifiers were trained on the TF-IDF features. A 5-Fold Cross-Validation strategy was used on the training set to ensure robustness.
- **Transformer Model:** A pre-trained distilbert-base-uncased model was fine-tuned. The training process was managed by a custom Hugging Face Trainer configured with a weighted loss function to handle class imbalance and an early stopping mechanism to prevent overfitting. A standard train/validation/test split was used instead of k-fold cross-validation due to the high computational cost of training large Transformer models.

4.7 Model Evaluation

The final phase of the methodology was to rigorously evaluate the performance of the trained models. This step is crucial for objectively assessing how well each model learned to generalize from the training data to new, unseen examples. The evaluation was conducted on the hold-out test set that was reserved for this final assessment.

To provide a comprehensive and fair comparison, a set of standard classification metrics was used:

- **Accuracy:** The overall percentage of reviews that were correctly classified.
- **Precision:** For each sentiment class, this measures the proportion of correct predictions among all predictions made for that class.
- **Recall (Sensitivity):** For each sentiment class, this measures the model's ability to correctly identify all actual instances of that class.
- **F1-Score:** The harmonic mean of precision and recall, providing a single score that balances both metrics.

4.8 Negative Reviews Factor Extraction

The second part of this project focused on uncovering the underlying reasons for negative sentiment expressed in drug reviews. Rather than stopping at sentiment classification, this section aimed to explain *why* patients were dissatisfied. By analyzing only the reviews labeled as negative (corresponding to a rating ≤ 3), several Natural Language Processing (NLP) techniques were applied to extract meaningful patterns and thematic insights.

- **Filtering Negative Reviews:** A filtered dataset was created by isolating reviews with the sentiment label 'negative'. These texts formed the foundation for the subsequent analyses, ensuring that only the most critical feedback was being studied.
- **Phrase Extraction with TextBlob:** To explore the specific issues raised in the reviews, noun phrases were extracted using the TextBlob library. This helped identify multi-word expressions that pointed to recurring complaints. Examples included side effects, weight gain, back pain, and mood swings. These phrases added context that single-word analysis might miss and highlighted common experiences shared by multiple users.
- **TF-IDF Vectorization:** To determine which words were both frequent and distinctive to negative reviews, TF-IDF (Term Frequency-Inverse Document Frequency) was applied. This helped identify high-impact terms such as pain, dizziness, nausea, vomiting, and severe. Unlike simple frequency counts, TF-IDF gave priority to words that were unusually common in negative texts compared to others.
- **N-Gram and Phrase Pattern Analysis:** Beyond individual terms, bigrams and trigrams were extracted to uncover commonly paired words that form more descriptive complaint structures. Phrases such as side effects, birth control, weight gain, and makes me sick revealed more precise concerns and allowed for better thematic categorization.
- **Topic Modeling with LDA:** Latent Dirichlet Allocation (LDA) was used to identify hidden thematic structures within the negative reviews. This unsupervised machine learning technique grouped words into coherent topics, each representing a common complaint theme. The five main topics revealed:

- Physical symptoms (e.g., pain, nausea, dizziness)
- Hormonal and menstrual side effects (e.g., period, bleeding, birth control)
- Mental and emotional impact (e.g., anxiety, depression, mood swings)
- Ineffectiveness of medication (e.g., didn't work, stopped, worse)
- Disruptions to daily life (e.g., sleep, fatigue, work performance)

Each topic provided a clearer understanding of what patients struggle with, going beyond numerical ratings to highlight the lived experiences behind them.

5. Sentiment Analysis

The central analytical task of this project was sentiment analysis. The objective of this phase was to apply the models developed in the methodology section to classify each drug review into one of the three predefined categories: positive, neutral, or negative.

To achieve this, the three trained models—Support Vector Machine, Logistic Regression, and the fine-tuned DistilBERT Transformer—were each tasked with predicting the sentiment of every review in the unseen hold-out test set. The primary goal of this analysis was to systematically compare the predictive power of the traditional machine learning approaches against the deep learning Transformer model. The effectiveness of each model was measured using a suite of standard classification metrics, with a particular focus on the weighted F1-score to account for the dataset's class imbalance. This comparative analysis is essential for identifying the most robust and accurate model architecture for this specific text classification problem. The detailed outcomes of this evaluation are presented in Section 7.

6. Transformer/ Deep Learning Models

The deep learning component of this study centered on the implementation of a Transformer model, a state-of-the-art architecture for natural language processing tasks. The specific model chosen was **distilbert-base-uncased**, a lighter and more efficient variant of the original BERT model, which maintains strong performance while requiring fewer computational resources.

The fundamental advantage of the Transformer architecture over traditional models lies in its ability to understand context. Unlike the TF-IDF approach, which treats a review as an unordered "bag-of-words," a Transformer processes the entire sequence of words. This allows it to learn the complex relationships between them, capturing nuances in meaning related to word order, negation (e.g., "not effective"), and sarcasm that are often missed by frequency-based methods.

The pre-trained DistilBERT model, already possessing a general understanding of the English language, was specifically adapted for this project's task through fine-tuning. This involved a carefully managed training strategy to maximize its effectiveness:

- **Handling Class Imbalance:** A custom WeightedLossTrainer was used to apply a weighted loss function during training. This forced the model to pay more attention to the underrepresented sentiment classes, preventing it from simply defaulting to the majority class.
- **Preventing Overfitting:** An **Early Stopping** mechanism was implemented. This monitored the model's performance on a separate validation set after each training epoch and automatically halted the process if performance did not improve. This disciplined approach ensures the model generalizes well to new, unseen data rather than memorizing the training set.

By leveraging a sophisticated architecture and a robust training strategy, this deep learning approach was positioned to achieve a high level of accuracy and contextual understanding. The detailed performance results are presented in the following section.

7. Results & Visualization

This section presents the insights and findings from the opinion mining, ABSA, modelling and negative review analysis.

7.1 Results for opinion mining, ABSA, modelling

- **Opinion Mining**

The rule-based OpinionMiner was applied to a sample of 30,000 reviews to extract specific opinion-target pairs. The analysis yielded significant insights into the primary topics of discussion within patient feedback.

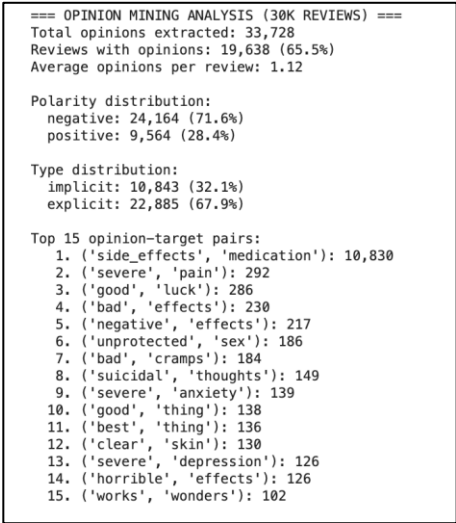


Figure 7.1.1: Opinion Mining Findings

Overall Statistics: The miner successfully extracted a total of 33,728 opinions from 19,638 unique reviews, indicating that opinions were found in 65.5% of the analyzed text. The sentiment of these extracted opinions was overwhelmingly negative, with 71.6% of all opinions being negative compared to just 28.4% positive. This highlights that patients are more likely to express specific, targeted opinions when they have a negative experience. The analysis also revealed that 67.9% of the opinions were explicit (based on lexicon words), while 32.1% were implicit (based on contextual patterns).

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=====
TOP OPINION SELECTIONS
=====

Top 10 Positive Opinions:
1. 'good' - 1,790 times
2. 'great' - 1,074 times
3. 'best' - 874 times
4. 'huge' - 473 times
5. 'love' - 420 times
6. 'positive' - 388 times
7. 'better' - 377 times
8. 'free' - 289 times
9. 'clear' - 257 times
10. 'healthy' - 229 times

Top 10 Negative Opinions:
1. 'side_effects' - 10,830 times
2. 'bad' - 2,271 times
3. 'severe' - 2,232 times
4. 'horrible' - 1,075 times
5. 'worst' - 914 times
6. 'terrible' - 787 times
7. 'negative' - 616 times
8. 'painful' - 434 times
9. 'awful' - 288 times
10. 'sore' - 225 times

Top 10 Opinion Targets:
1. 'medication' - 11,208 times
2. 'effects' - 1,127 times
3. 'pain' - 691 times
4. 'cramps' - 612 times
5. 'experience' - 586 times
6. 'thing' - 573 times
7. 'medicine' - 391 times
8. 'acne' - 378 times
9. 'anxiety' - 304 times
10. 'luck' - 302 times

Top 5 Explicit Opinions:
1. 'bad' - 2,310 times
2. 'severe' - 2,240 times
3. 'good' - 1,800 times
4. 'great' - 1,077 times
5. 'horrible' - 1,076 times

Top 5 Implicit Opinions:
1. 'side_effects' - 10,830 times
2. 'symptoms_improved' - 13 times

TOP RESULTS:
Top positive opinion: ('good', 1790)
Top negative opinion: ('side_effects', 10830)
Most targeted aspect: ('medication', 11208)

```

Figure 7.1.2: Opinion Mining Findings

The most striking finding is the overwhelming dominance of a single theme.

- **Top Opinion-Target Pair:** The pair ('side_effects', 'medication') was extracted 10,830 times. This single pair accounts for nearly a third of all opinions found and is almost 40 times more frequent than the next most common pair, demonstrating that side effects are the single most critical topic of discussion.
- **Top Negative Opinions:** The term "side_effects" was also the most frequent negative opinion word, appearing over 10,000 times. This was followed by expected negative terms like "bad" (2,271 times) and "severe" (2,232 times).
- **Top Positive Opinions:** Positive feedback was more general, with words like "good" (1,790 times), "great" (1,074 times), and "best" (874 times) leading the list.

- **Top Opinion Targets:** As expected, "medication" was the most frequently discussed target (11,208 mentions), followed by related terms like "effects," "pain," and "cramps."

In summary, the opinion mining results clearly show that while general positive sentiment exists, specific, targeted feedback is dominated by negative discussions centered almost entirely on the side effects of medication.

- **ABSA**

=== DISCOVERED ASPECTS (Top 20) - WITH NEUTRAL ===		
1. 'my doctor' - 3409 mentions	Positive: 28.6%	Negative: 30.1% Neutral: 41.3%
2. 'my period' - 2711 mentions	Positive: 29.9%	Negative: 33.3% Neutral: 36.7%
3. 'my life' - 2520 mentions	Positive: 40.2%	Negative: 32.4% Neutral: 27.4%
4. 'this medication' - 2475 mentions	Positive: 38.0%	Negative: 32.6% Neutral: 29.4%
5. 'this medicine' - 2026 mentions	Positive: 44.0%	Negative: 29.3% Neutral: 26.7%
6. 'the pill' - 1761 mentions	Positive: 32.0%	Negative: 35.0% Neutral: 33.0%
7. 'side effects' - 1728 mentions	Positive: 27.9%	Negative: 40.4% Neutral: 31.6%
8. 'this pill' - 1622 mentions	Positive: 38.9%	Negative: 31.6% Neutral: 29.5%
9. 'the side effects' - 1504 mentions	Positive: 30.0%	Negative: 43.5% Neutral: 26.5%
10. 'a lot' - 1377 mentions	Positive: 31.9%	Negative: 39.1% Neutral: 29.0%
11. 'the pain' - 1323 mentions	Positive: 10.6%	Negative: 85.1% Neutral: 4.3%
12. 'this drug' - 1312 mentions	Positive: 36.9%	Negative: 34.7% Neutral: 28.4%
13. 'my skin' - 1233 mentions	Positive: 51.5%	Negative: 29.2% Neutral: 19.3%
14. 'birth control' - 1122 mentions	Positive: 39.1%	Negative: 29.5% Neutral: 31.4%
15. 'no side effects' - 1112 mentions	Positive: 23.4%	Negative: 71.5% Neutral: 5.1%
16. 'the doctor' - 1037 mentions	Positive: 27.4%	Negative: 31.2% Neutral: 41.4%
17. 'my body' - 991 mentions	Positive: 32.7%	Negative: 40.0% Neutral: 27.4%
18. 'my face' - 982 mentions	Positive: 36.3%	Negative: 39.1% Neutral: 24.5%
19. 'the day' - 911 mentions	Positive: 32.0%	Negative: 35.8% Neutral: 32.2%
20. 'the medication' - 903 mentions	Positive: 31.7%	Negative: 39.5% Neutral: 28.8%

Figure 7.1.3: ABSA Results

The analysis of the top 20 most frequently mentioned aspects (among the 30,000 samples taken) reveals clear patterns in patient feedback:

- **Strongly Negative Aspects:** Certain topics were overwhelmingly associated with negative sentiment.

- "The pain" stands out as the most negative aspect, with 85.1% of its 1,323 mentions being negative. This indicates that when pain is discussed, it is almost always in a negative context.
- "Side effects" and "the side effects" also skewed heavily negative (40.4% and 43.5% respectively), reinforcing the findings from the rule-based miner that adverse effects are a primary concern.
- **Strongly Positive Aspects:** Conversely, some aspects were clearly linked to positive outcomes.
 - "My skin" was the most positive topic, with 51.5% of its 1,233 mentions being positive. This strongly suggests successful treatment for dermatological conditions like acne.
 - General terms like "this medicine" (44.0% positive) and "my life" (40.2% positive) also showed high positive sentiment, indicating overall satisfaction and improved quality of life for many users.
- **Neutral and Factual Aspects:** Some of the most common topics were surprisingly neutral.
 - "My doctor" and "the doctor" were the most frequently mentioned aspects overall, but also the most neutral (41.3% and 41.4% neutral, respectively). This suggests that patients often mention their doctor in a factual context (e.g., "my doctor prescribed...") rather than expressing strong sentiment about them.
 - Anomalous Finding: An interesting and counter-intuitive result was observed for the aspect "no side effects". Despite its positive phrasing, it was associated with 71.5% negative sentiment. This is likely due to complex sentence structures where VADER misinterprets the context, such as "I had no side effects at first, but then things got worse," or "This drug is useless, but at least there are no side effects." This highlights a limitation of sentence-level sentiment analysis for complex phrases.

In summary, the ABSA approach successfully identified and quantified sentiment towards a wide range of topics, from specific symptoms like "pain" to outcomes like "my skin," providing a more nuanced and comprehensive view of the patient experience.

- **Model Performances**

This section presents the final performance evaluation of the three sentiment analysis models. Each model was assessed on the hold-out test set, ensuring an unbiased measure of its ability to generalize to new data.

The results, summarized in the tables below, highlight the performance differences between the traditional machine learning and transformer based approaches.

- **Support Vector Machine (SVM)**

The SVM model achieved a final accuracy of 76.0%. The detailed metrics show a strong performance on the "Positive" class but significantly lower precision and recall for the "Neutral" class.

SVM Test Results:				
	precision	recall	f1-score	support
negative	0.67	0.71	0.69	7966
neutral	0.26	0.27	0.27	2852
positive	0.87	0.84	0.86	21262
accuracy			0.76	32080
macro avg	0.60	0.61	0.60	32080
weighted avg	0.77	0.76	0.76	32080

Figure 7.1.4: SVM Model Performance

- **Logistic Regression (LR)**

The Logistic Regression model achieved a final accuracy of 76.6%, slightly outperforming the SVM. Its performance profile is similar, with high scores for the "Positive" class and weaker results for the "Neutral" class.

LR Test Results:				
	precision	recall	f1-score	support
negative	0.68	0.71	0.69	7966
neutral	0.27	0.25	0.26	2852
positive	0.86	0.86	0.86	21262
accuracy			0.77	32080
macro avg	0.60	0.60	0.60	32080
weighted avg	0.76	0.77	0.76	32080

Figure 7.1.5: LR Model Performance

- **DistilBert Transformer**

The fine-tuning process for the DistilBERT model was carefully monitored to ensure the selection of the best-performing model and to prevent overfitting. The training logs provide a clear picture of this process. The model was trained for a potential of five epochs, with its performance evaluated against a validation set after each epoch.

Epoch	Training Loss	Validation Loss	Accuracy	F1	Precision	Recall
1	0.713300	0.691403	0.771717	0.790180	0.822261	0.771717
2	0.611900	0.667168	0.786869	0.810683	0.848666	0.786869
3	0.471000	0.795201	0.819596	0.828314	0.840299	0.819596
4	0.368900	0.911334	0.813737	0.828822	0.850486	0.813737

Figure 7.1.6: DistilBert Performance

As shown in Figure 7.1.6, the Training Loss consistently decreased, which indicates the model was learning from the training data. However, the crucial metric is the Validation Loss. It decreased from Epoch 1 to Epoch 2, reaching its minimum value of 0.6672. After this point, the Validation Loss began to rise, which is a classic sign of overfitting—the model was starting to memorize the training data rather than generalizing. The EarlyStopping callback correctly identified this trend and automatically halted the training process, selecting the model from Epoch 2 as the optimal version.

This best model from Epoch 2 was then evaluated on the completely unseen hold-out test set. The final performance metrics were:

- Final Accuracy: 78.1%
- Final Weighted F1-Score: 80.5%
- Final Weighted Precision: 84.4%
- Final Weighted Recall: 78.1%

In summary, the disciplined training strategy successfully identified the point of peak performance, preventing overfitting and ensuring that the final reported results reflect the model's true ability to generalize to new data.

The final F1-score of 80.5% demonstrates a strong and balanced performance across all sentiment classes.

7.2 Negative Review Analysis

In order to provide explanation for the negative sentiment classifications, this project performs a detailed text analysis on all negative reviews. The results are as summarized:

1. Top keywords (Word Frequency)

One of the first steps involved identifying which words appeared most frequently in negative reviews. Using the CountVectorizer tool from the scikit-learn library, the review texts were analyzed and converted into a frequency distribution.

The most frequently mentioned terms included:

- *pain, effects, feel, doctor, months, and day*

These words reflect a mix of physical symptoms (like **pain**), general feedback on experience (**feel, effects**), and references to healthcare or time duration (**doctor, months, day**). While some words appear neutral on the surface, their context within the reviews often points to dissatisfaction or discomfort.

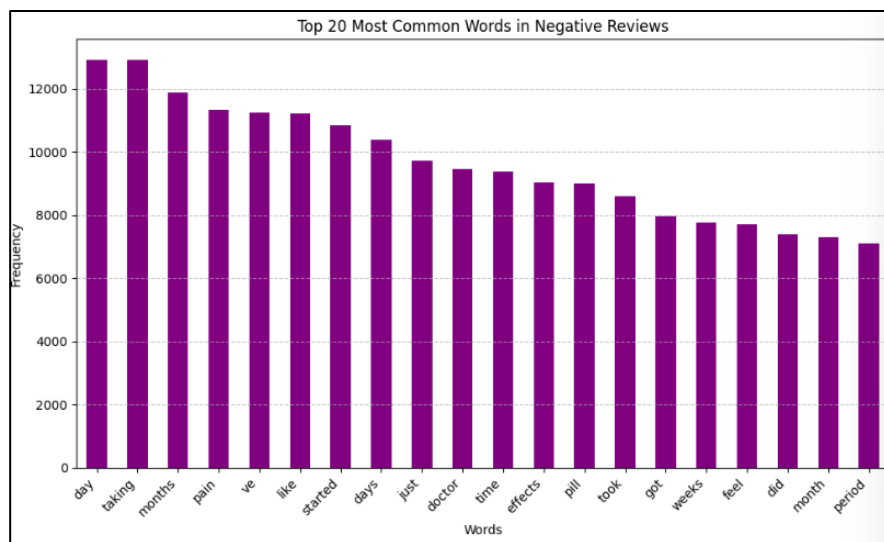


Figure 7.2.1: Top 20 Most Common Words in Negative Reviews

2. Top TF-IDF Terms

While the raw word frequency shows which terms appear most often, TF-IDF helps us identify which words are most unique and important within negative reviews.

The bar chart below shows the top 20 terms ranked by their TF-IDF scores:

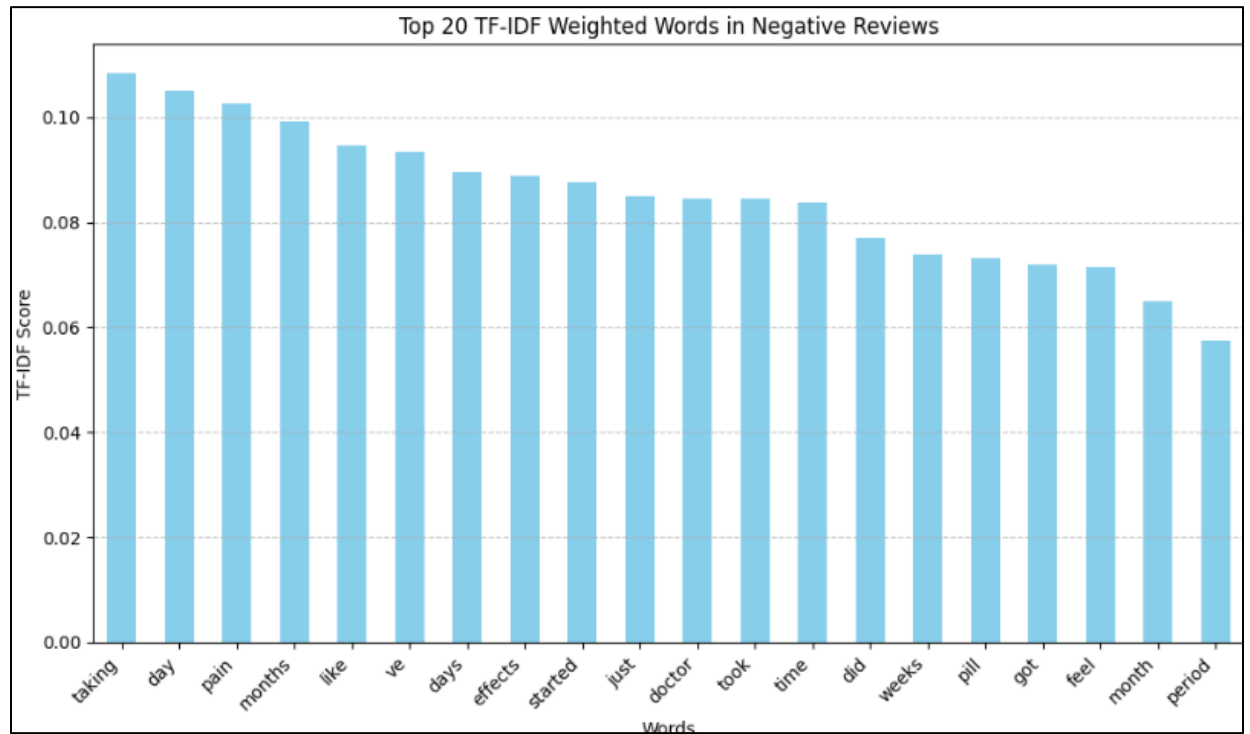


Figure 7.2.2: Top 20 TF-IDF Weighted Words in Negative Reviews

Some of these terms, such as "pain", "effects", "doctor", and "started", represent core complaint contexts — related to physical symptoms, medication experiences, and interactions with healthcare providers. These words are not just frequent but also distinctive to dissatisfaction.

3. Common Complaint Phrases (Bigrams)

To identify more specific complaints, we extracted bigrams which are two-word phrases that often reflect more distinct issues. The chart below displays the top 20 bigrams in the negative reviews.

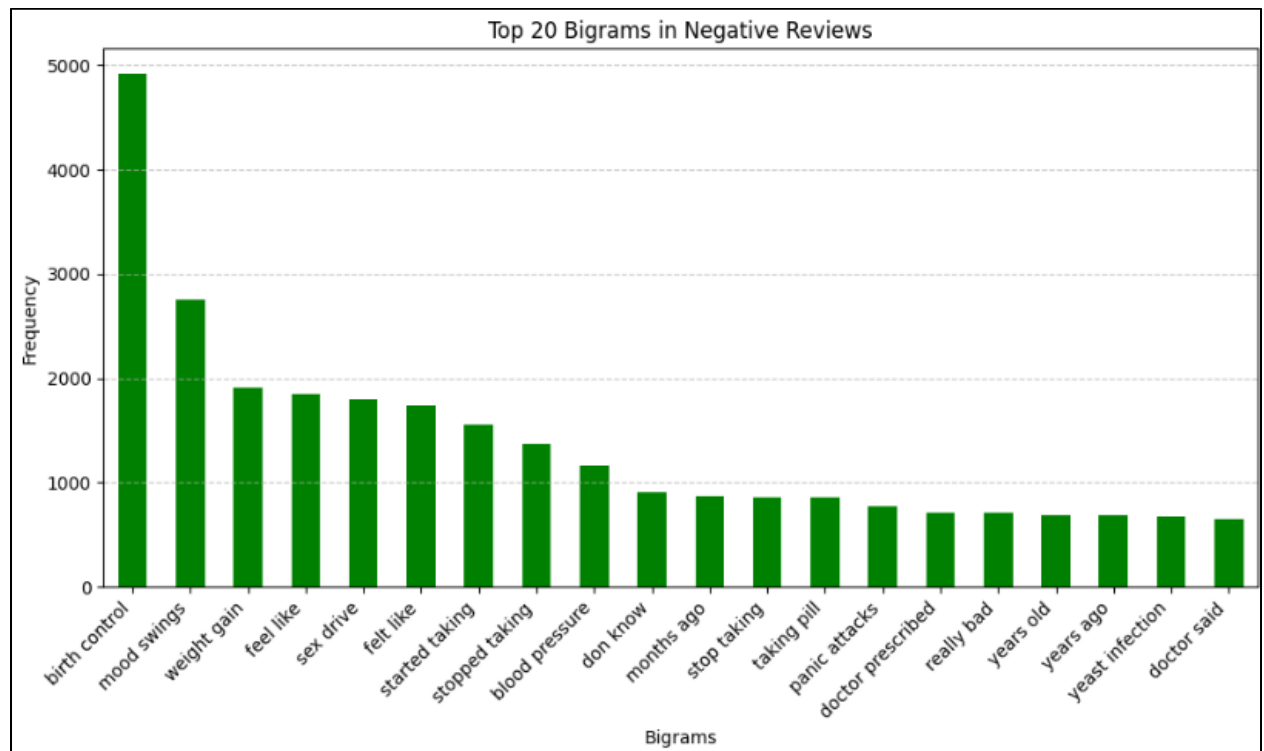


Figure 7.2.3: Top 20 Bigrams in Negative Reviews

These frequent phrases offer clearer insight into what patients are experiencing. For example:

- **"birth control"** and **"weight gain"** point to hormonal or medication-related issues.
- **"mood swings"**, **"panic attacks"**, and **"sex drive"** suggest emotional and psychological side effects.
- Terms like **"doctor prescribed"**, **"stop taking"**, and **"really bad"** reflect actions or evaluations tied to the user's experience.

This level of phrase-based analysis goes beyond individual keywords and captures *meaningful medical narratives* in patient language.

4. TF-IDF Weighted Bigrams

While raw frequency highlights the most commonly occurring bigrams, TF-IDF weighting emphasizes phrases that are uniquely significant in negative reviews.

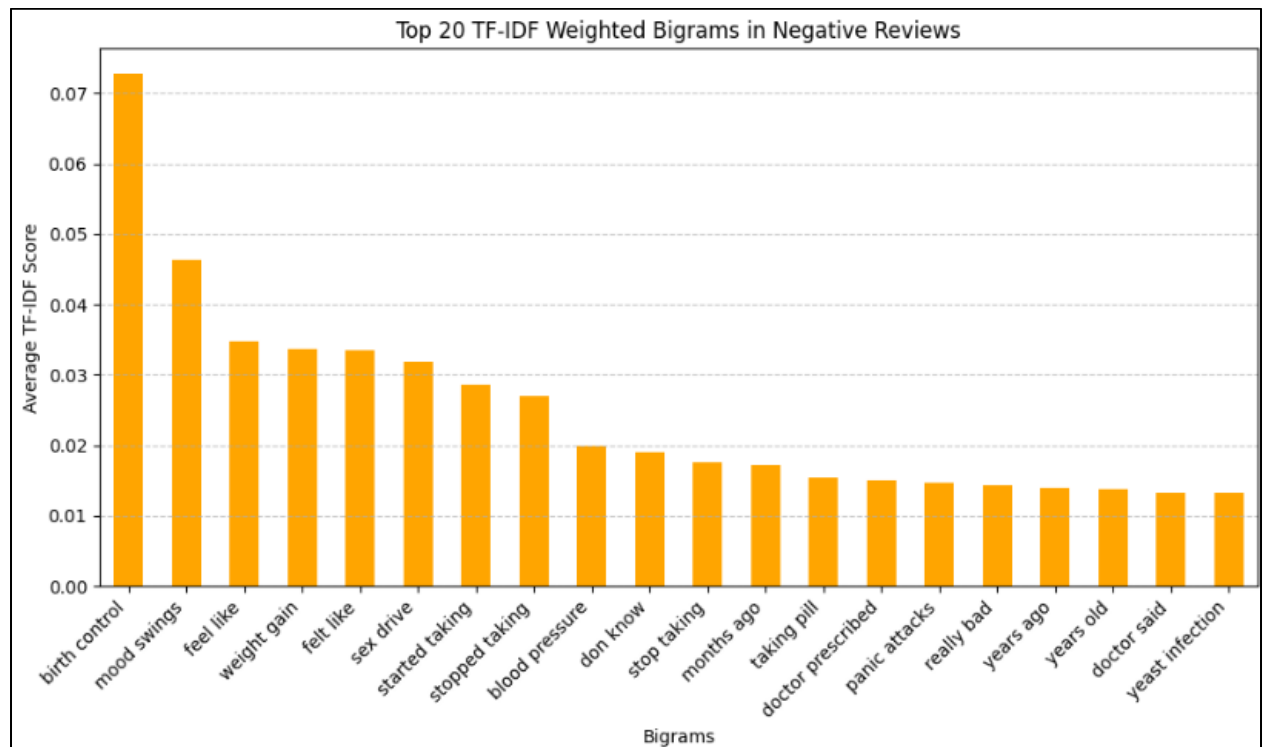


Figure 7.2.4: Top 20 TF-IDF Weighted Bigrams in Negative Reviews

This analysis reinforced earlier findings while de-emphasizing overly generic phrases.

For example:

- **"birth control"** and **"mood swings"** remain highly weighted, showing they are not only common but also central to negative sentiment.
- **"felt like"**, **"sex drive"**, and **"panic attacks"** highlight emotional and physiological side effects that may not surface in a pure frequency count.

These TF-IDF-weighted phrases bring attention to **context-rich complaints** that are strongly associated with dissatisfaction and not just volume.

5. Trigram Analysis

Trigrams — three-word phrases — provide even greater context into the structure of user complaints. These combinations help reveal not only the issue but often the **tone** or **severity** of the experience.

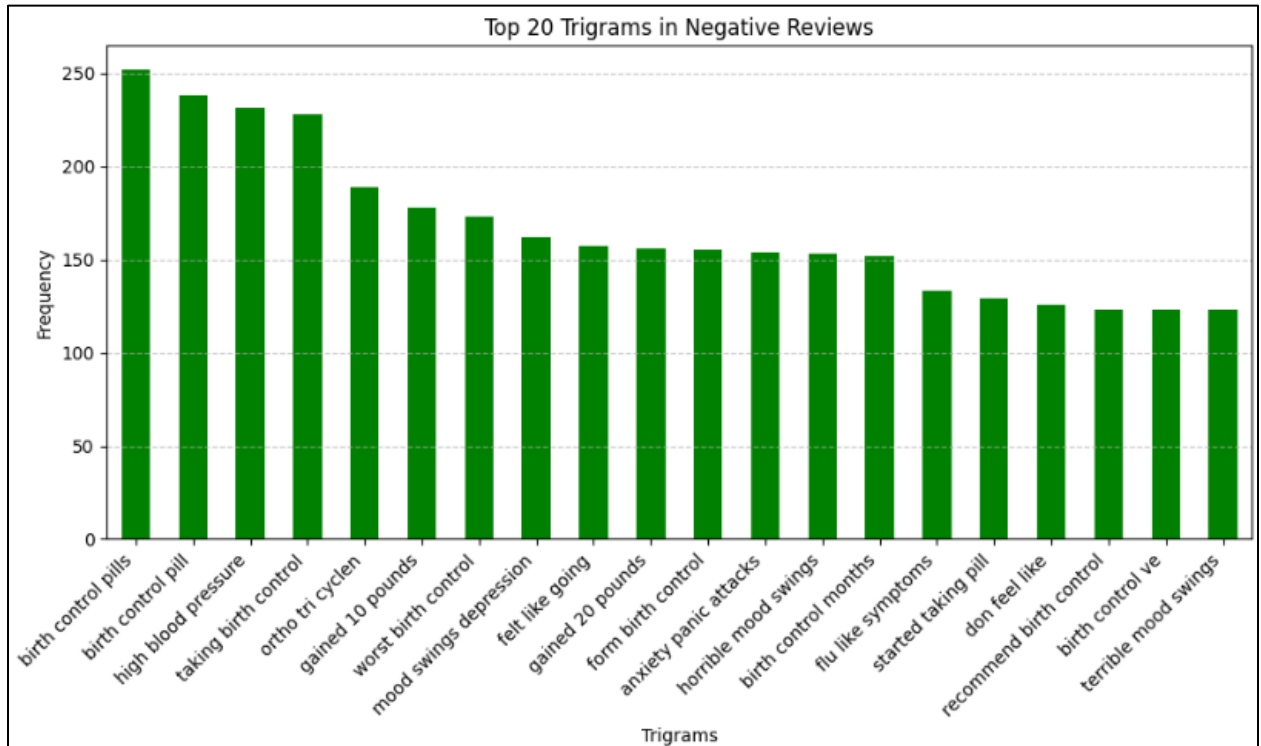


Figure 7.2.5: Top 20 Trigrams in Negative Reviews

Examples like:

- **"high blood pressure"**, **"birth control pills"**, and **"started taking pill"** reflect medical conditions or treatment progressions
- **"gained 10 pounds"**, **"feel like going"**, and **"worst birth control"** suggest side effects and dissatisfaction
- **"anxiety panic attacks"** and **"terrible mood swings"** point to emotional and psychological side effects that significantly impacted the user's experience

By capturing these longer expressions, trigram analysis reinforces and deepens the insights extracted from earlier bigram and word-level analysis.

6. Topic Modeling (LDA)

To uncover deeper thematic structures in the negative reviews, we used **Latent Dirichlet Allocation (LDA)** — an unsupervised topic modeling technique. It grouped reviews into latent topics based on word distributions.

```
Topic 1:
['effects', 'anxiety', 'taking', 'medication', 'drug', 'years', 'depression', 'months', 'started', 'doctor']

Topic 2:
['bleeding', 'period', 'months', 'got', 'month', 'days', 'weeks', 'shot', 'started', 'having']

Topic 3:
['pill', 've', 'months', 'control', 'weight', 'acne', 'birth', 'month', 'taking', 'started']

Topic 4:
['pain', 'day', 'taking', 'days', 'took', 'doctor', 'severe', 'stomach', 'started', 'medication']

Topic 5:
['like', 'day', 'night', 'just', 'sleep', 'work', 'feel', 'time', 'took', 'don']
```

Figure 7.2.6: Top 5 Complaint Topics Identified by LDA

The model identified 5 dominant topics:

- **Topic 1 – Mental/Emotional Effects:**
effects, anxiety, depression, medication, taking, doctor
- **Topic 2 – Menstrual & Hormonal Issues:**
bleeding, period, shot, weeks, months, having
- **Topic 3 – Birth Control Side Effects:**
pill, acne, weight, control, birth, taking
- **Topic 4 – Physical Reactions & Stomach Pain:**
pain, stomach, doctor, severe, days, started
- **Topic 5 – Fatigue & Sleep Disruption:**
sleep, night, work, feel, time, don't

These topics reflect **distinct dissatisfaction categories**, ranging from mental health to physical discomfort and hormonal disruption. LDA helps generalize themes that TF-IDF or phrase extraction may isolate.

Visualizing Complaint Topics using Word Clouds

To enhance interpretability of the LDA topic modeling, word clouds were generated for each of the five topics. Each cloud displays the most prominent words associated with that complaint cluster.

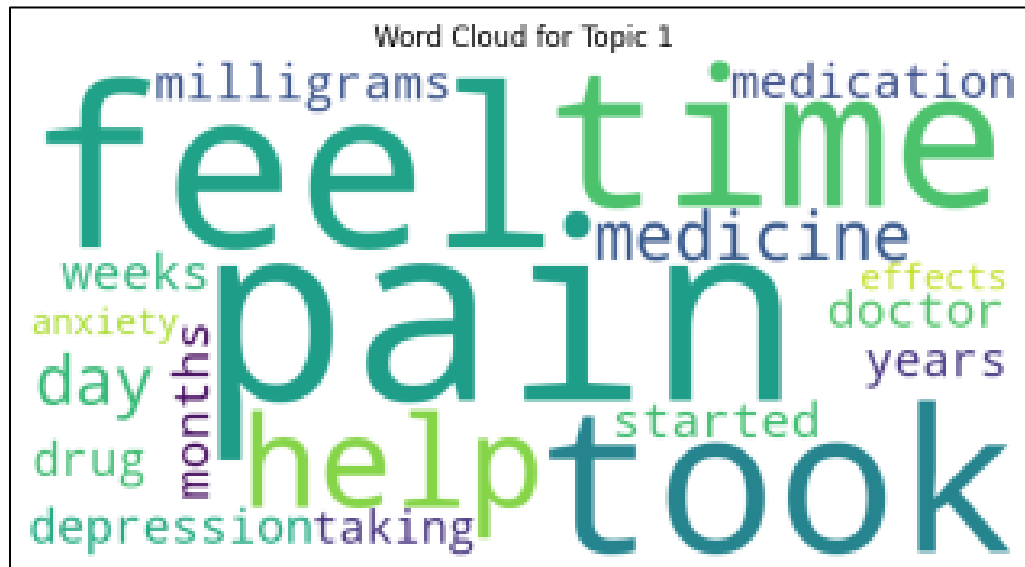


Figure 7.2.7: Word Cloud for Topic 1 – Mental/Physical Discomfort

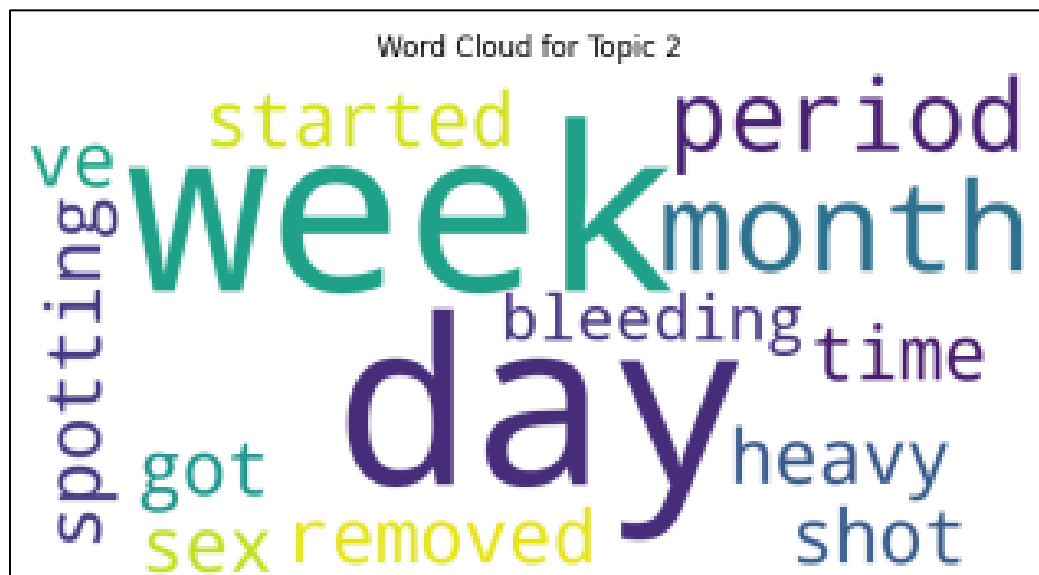
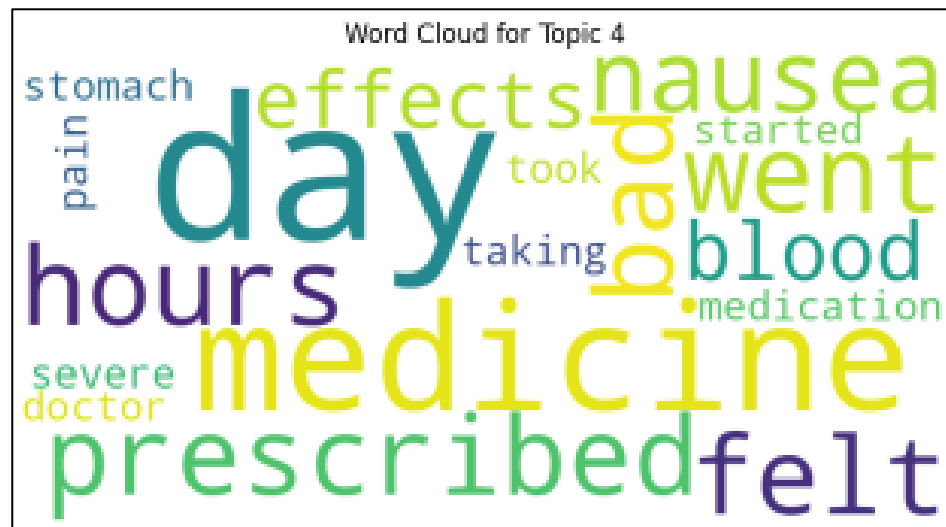
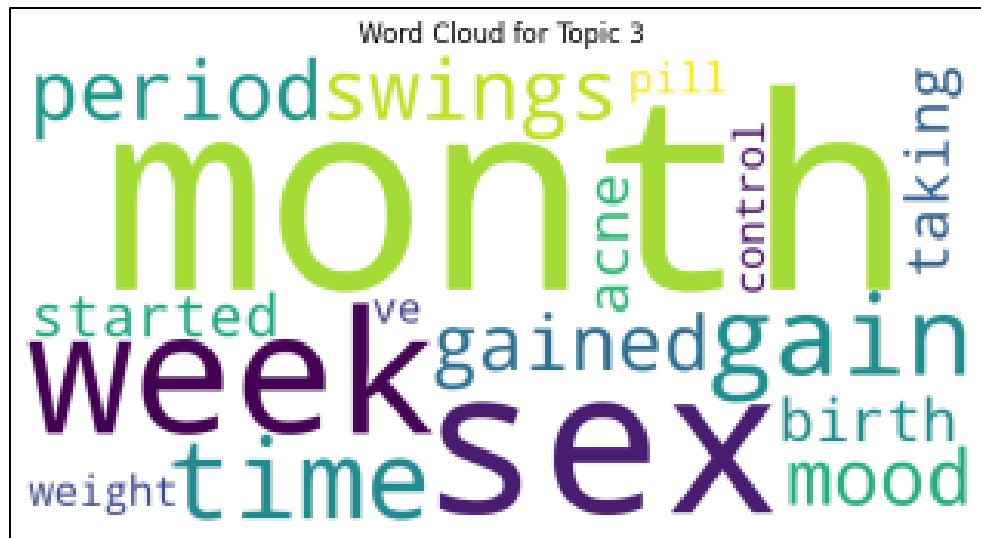


Figure 7.2.8: Word Cloud for Topic 2 – Menstrual Irregularities



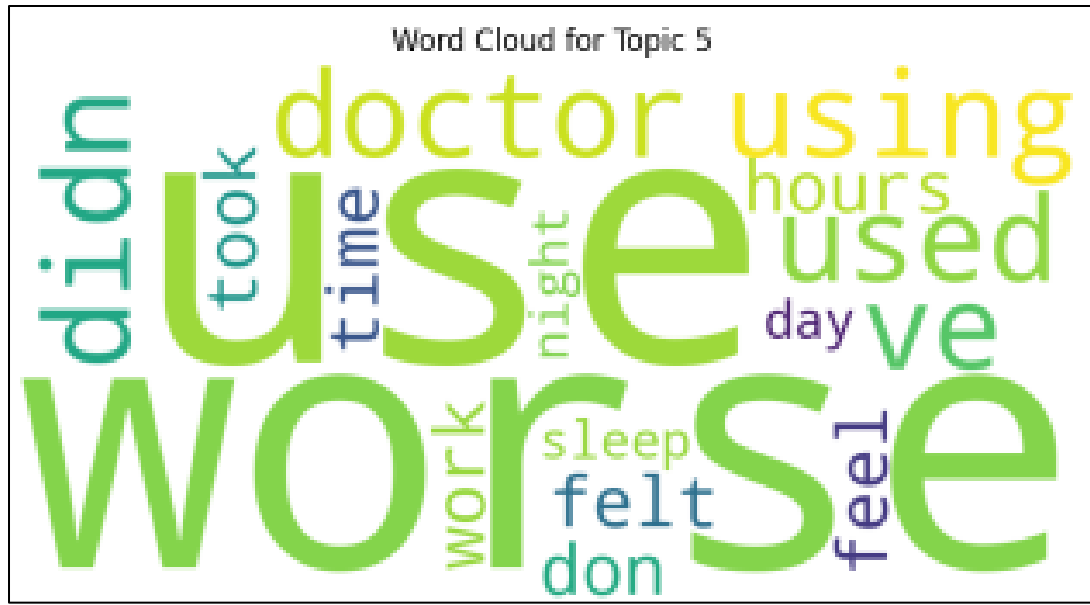


Figure 7.2.11: Word Cloud for Topic 5 – Worsening Condition and Fatigue

7. Sentiment Polarity Distribution

To better understand the severity and nuance of negativity in the reviews, we plotted the **sentiment polarity scores** of all texts labeled as negative. These scores range from -1 (extremely negative) to +1 (positive), with 0 being neutral.

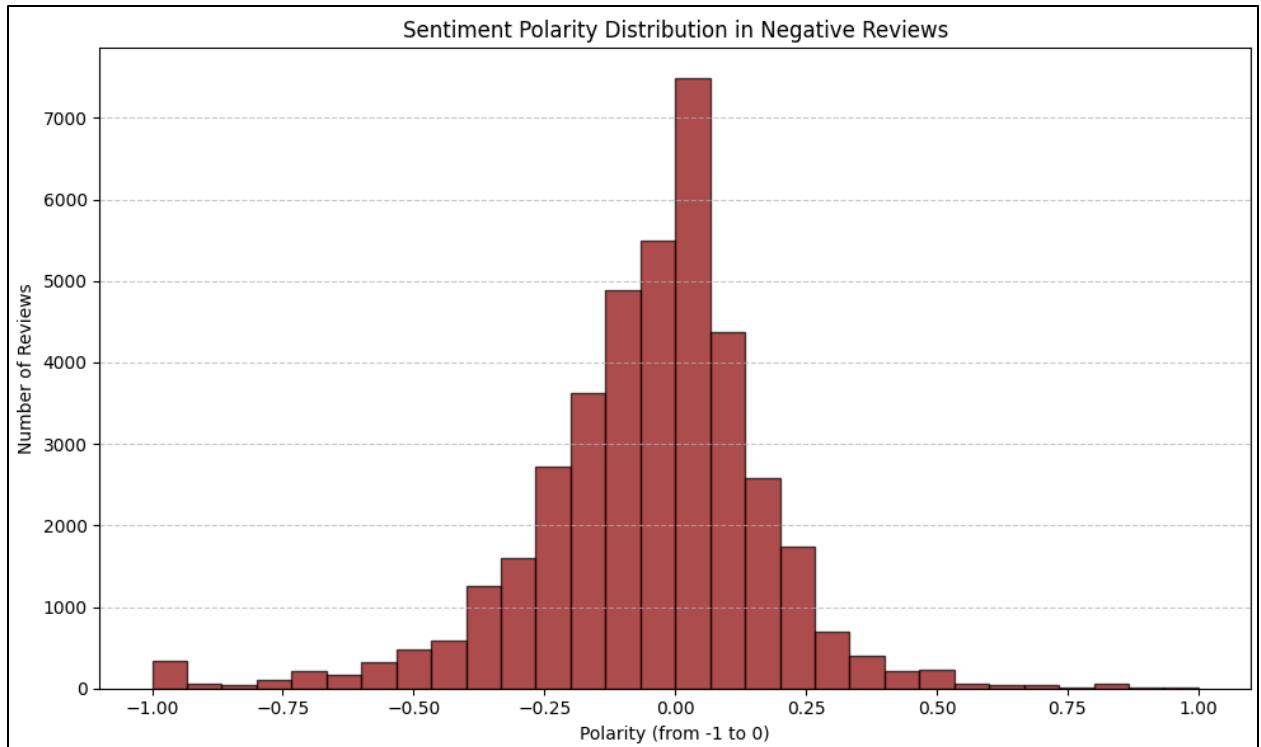


Figure 7.2.12: Distribution of Polarity Scores in Negative Reviews

The histogram shows that while most reviews cluster around the **mildly negative range (-0.2 to 0)**, a significant number dip toward **-0.5 and beyond**, indicating intense dissatisfaction. Only a small fraction of “negative” reviews contain mixed or slightly positive phrases, which is expected due to the complexity of language.

This confirms that the classification threshold used was accurate in capturing real sentiment and not just arbitrary numeric ratings.

8. Discussion

This study explored how sentiment analysis and text mining can be applied to real-world drug review data to extract both classifications and deeper insights. While traditional models such as Logistic Regression and Support Vector Machines were effective in predicting sentiment labels, they struggled with understanding more complex patterns and subtle cues in patient language. The use of a fine-tuned DistilBERT model proved useful in addressing these gaps, particularly in handling contextual relationships within longer review texts.

Beyond sentiment classification, the second part of the project focused on examining why certain reviews were negative. By isolating these reviews and analyzing them using methods like TF-IDF, n-gram extraction, and topic modeling, we were able to identify recurring concerns. These included side effects, weight changes, emotional reactions, and lack of treatment effectiveness. Word clouds and LDA topics helped to visualize these patterns clearly, providing a more grounded understanding of patient experiences.

The polarity distribution graph also supported the negative classification logic, showing that most reviews clustered around mildly to strongly negative scores. This confirmed that the sentiment labels were aligned with how users genuinely expressed their feedback.

There were some limitations throughout the analysis. Very short reviews offered little context and were sometimes misclassified. In addition, LDA topic labels had to be interpreted manually, and some results were vague without deeper domain knowledge. Another consideration is that we did not factor in metadata like drug type or patient condition, which could have added useful context.

Overall, combining sentiment classification with negative review analysis gave us both prediction accuracy and interpretability. This two-part structure helped move beyond simple classification and offered a clearer picture of user dissatisfaction in healthcare feedback.

9. Conclusion & Future Work

9.1 Conclusion

This project sets out to analyze patient drug reviews using a combination of machine learning and text mining techniques. The sentiment classification phase showed that both traditional and transformer-based models are capable of detecting positive, neutral, and negative sentiments with reasonable accuracy. More importantly, the second phase—focusing on the causes behind negative reviews—uncovered valuable themes such as side effects, emotional distress, and inefficacy of treatment. These findings offer real-world relevance, helping healthcare professionals and pharmaceutical companies better understand patient concerns and experiences beyond simple ratings.

9.2 Future Work

There are several ways this project could be expanded or refined in future iterations:

- **Incorporating additional metadata** such as medical conditions to relate sentiments to specific treatments.
- **Exploring more advanced NLP techniques**, including models like BERTweet, RoBERTa, or BERTopic, which may capture nuanced complaints more accurately.
- **Enhancing aspect-based sentiment analysis** to map sentiments to specific components (e.g., side effects vs. dosage).
- **Visualizing trends over time**, which could help identify long-term patterns in drug feedback.
- **Improving class balance techniques**, as negative reviews were less frequent and required oversampling strategies.

These directions would not only improve the accuracy of sentiment detection but also make the results more meaningful for real-world healthcare applications

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Appendix

```
from sklearn.model_selection import StratifiedKFold, train_test_split
from sklearn.svm import LinearSVC
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, accuracy_score
import pandas as pd
import numpy as np
import time

X_tfidf_full = tfidf_matrix_full
y_full = df2['sentiment_label1'].values

#clean data (safety check)
mask = ~pd.isna(y_full)
X_clean = X_tfidf_full[mask]
y_clean = y_full[mask]

#80% for training/CV, 20% for the final unseen test
X_train_main, X_test_final, y_train_main, y_test_final = train_test_split(
    X_clean, y_clean,
    test_size=0.2,
    random_state=42,
    stratify=y_clean
)

print(f"Main Training Set Size: {X_train_main.shape[0]:,} samples")
print(f"Final Hold-Out Test Set Size: {X_test_final.shape[0]:,} samples")
```

Main Training Set Size: 128,318 samples
Final Hold-Out Test Set Size: 32,080 samples

Figure 1: Train-Test Split on Data

```

: #LR: Training with K-Fold Cross-Validation
print("\n LR: Running 5-Fold Cross-Validation for model evaluation...")
lr_metrics = []
start_time_lr = time.time()

for fold, (train_idx, val_idx) in enumerate(skf.split(X_train_main, y_train_main)):
    print(f"LR CV Fold {fold + 1}/5...")
    X_train_fold, X_val_fold = X_train_main[train_idx], X_train_main[val_idx]
    y_train_fold, y_val_fold = y_train_main[train_idx], y_train_main[val_idx]

    lr = LogisticRegression(max_iter=1000, class_weight='balanced', solver='liblinear')
    lr.fit(X_train_fold, y_train_fold)
    y_pred_val = lr.predict(X_val_fold)
    lr_metrics.append(accuracy_score(y_val_fold, y_pred_val))

print(f"LR Cross-Validation finished in {(time.time() - start_time_lr)/60:.1f} minutes.")
print(f"LR Average CV Accuracy: {np.mean(lr_metrics):.4f} (± {np.std(lr_metrics):.4f})")

#final training and testing
final_lr = LogisticRegression(max_iter=1000, class_weight='balanced', solver='liblinear')
final_lr.fit(X_train_main, y_train_main)

y_pred_lr_final = final_lr.predict(X_test_final)

print("\nLR Test Results:")
print(classification_report(y_test_final, y_pred_lr_final))

```

LR: Running 5-Fold Cross-Validation for model evaluation...
 LR CV Fold 1/5...
 LR CV Fold 2/5...
 LR CV Fold 3/5...
 LR CV Fold 4/5...
 LR CV Fold 5/5...
 LR Cross-Validation finished in 0.2 minutes.
 LR Average CV Accuracy: 0.7675 (± 0.0018)

Figure 2: LR Model Training

```

: #SVM: Training with K-Fold Cross-Validation

skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
svm_metrics = []
start_time_svm = time.time()

for fold, (train_idx, val_idx) in enumerate(skf.split(X_train_main, y_train_main)):
    print(f"SVM CV Fold {fold + 1}/5...")
    X_train_fold, X_val_fold = X_train_main[train_idx], X_train_main[val_idx]
    y_train_fold, y_val_fold = y_train_main[train_idx], y_train_main[val_idx]

    svm = LinearSVC(class_weight='balanced')
    svm.fit(X_train_fold, y_train_fold)
    y_pred_val = svm.predict(X_val_fold)
    svm_metrics.append(accuracy_score(y_val_fold, y_pred_val))

print(f"SVM Cross-Validation finished in {(time.time() - start_time_svm)/60:.1f} minutes.")
print(f"SVM Average CV Accuracy: {np.mean(svm_metrics):.4f} (± {np.std(svm_metrics):.4f})")

#final training and testing
final_svm = LinearSVC(class_weight='balanced')
final_svm.fit(X_train_main, y_train_main)

y_pred_svm_final = final_svm.predict(X_test_final)

print("\nSVM Test Results:")
print(classification_report(y_test_final, y_pred_svm_final))

```

SVM CV Fold 1/5...
 SVM CV Fold 2/5...
 SVM CV Fold 3/5...
 SVM CV Fold 4/5...
 SVM CV Fold 5/5...
 SVM Cross-Validation finished in 0.4 minutes.
 SVM Average CV Accuracy: 0.7588 (± 0.0016)

Figure 3: SVM Model Training

```

:
#model initialization and training setup
model = DistilBertForSequenceClassification.from_pretrained('distilbert-base-uncased', num_labels=3)

training_args = TrainingArguments(
    output_dir="./results",
    num_train_epochs=5,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    learning_rate=2e-5,
    weight_decay=0.01,
    warmup_ratio=0.1,
    eval_strategy="epoch",
    save_strategy="epoch",
    load_best_model_at_end=True,
    metric_for_best_model="eval_loss",
    greater_is_better=False,
    report_to="none"
)

trainer = WeightedLossTrainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=val_dataset,
    compute_metrics=compute_metrics,
    tokenizer=tokenizer,
    callbacks=[EarlyStoppingCallback(early_stopping_patience=2)]
)

trainer.train()

#testing the model on the final test set
test_results = trainer.evaluate(test_dataset)
print("\nFinal Test Set Performance:")
print(test_results)

```

Figure 4: DistilBert Model Training Setup